

STAT 709: Homework 4

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1. Textbook exercises

Jun Shao's *Mathematical Statistics*, Chapter 2: Exercise 31 (required); Exercise 25 (optional).

2. Prediction

Give an example in which Z determines Y exactly, but $\text{Var}(Z | Y) = \text{Var}(Z)$ almost surely. Require Y to be nonconstant and Z to have finite, positive variance. Verify both properties.

3. Exponential families

Let $X \in \mathbb{R}^d$ have a density, with respect to Lebesgue measure, of the form

$$f_\theta(x) = h(x) \exp\{\theta^\top T(x) - \xi(\theta)\}, \quad \theta \in \Theta,$$

Here h is nonnegative and Borel measurable, $T : \mathbb{R}^d \rightarrow \mathbb{R}^k$ is continuous, and the natural parameter space is

$$\Theta = \left\{ \theta \in \mathbb{R}^k : 0 < \int_{\mathbb{R}^d} h(x) e^{\theta^\top T(x)} dx < \infty \right\}.$$

Fix θ_0 in its interior, and let B be an open convex neighborhood of θ_0 contained in Θ .

(1) Show that

$$\xi(\theta) = \log \int h(x) e^{\theta^\top T(x)} dx$$

and that $\xi \in C^\infty(B)$. Thus, for every multi-index $\alpha = (\alpha_1, \dots, \alpha_k) \in \{0, 1, 2, \dots\}^k$, with $|\alpha| = \alpha_1 + \dots + \alpha_k$, $\partial^{|\alpha|} \xi / (\partial \theta_1^{\alpha_1} \dots \partial \theta_k^{\alpha_k})$ exists and is continuous on B .

(2) For $\theta \in B$, express $\mathbb{E}_\theta T(X)$, $\text{Cov}_\theta(T(X))$, and the moment generating function of $T(X)$ using ξ and its derivatives. Specify where the moment generating function is finite.

(3) Show that Θ is convex and that ξ is convex on Θ . Show that ξ is strictly convex on B if $T(X)$ is not supported on any proper affine hyperplane: there are no $a \neq 0$ and b with $a^\top T(X) = b$ almost surely under P_{θ_0} .

4. Bias–variance tradeoff

Let $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(\theta)$, where $\theta \in [0, 1]$, and write $\bar{X} = n^{-1} \sum_{i=1}^n X_i$. Consider the estimator

$$\hat{\theta} = (1 - \varepsilon_n) \left(\bar{X} - \frac{1}{2} \right) + \frac{1}{2}, \quad 0 < \varepsilon_n < 1.$$

The constant ε_n may depend only on n . Setting $\varepsilon_n = 0$ would recover the sample mean.

(1) Find the bias and variance of $\hat{\theta}$.

(2) Find the choice of $\varepsilon_n \in (0, 1)$ that minimizes the worst-case mean squared error $\sup_{\theta \in [0, 1]} \mathbb{E}_\theta[(\hat{\theta} - \theta)^2]$.